

# Portfolio

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Hatixhe Merkaj

Electronics and Communication engineering  
student

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# Who I Am

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HATIXHE MERKAJ

I am a third year electronics and communication engineering student at Istanbul Technical University

I take great interest in electronics, aviation systems, and hyperloop technologies, I also enjoy nature walks and good literature.



# What I Do

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HATIXHE MERKAJ

I have a great theoretical knowledge of circuit design and circuit analysis. I also have good programming skills in C, python and Matlab, and I have mastered the use of of Zemax optic studio through projects in the optical engineering spectrum. I also have experience in the control field thanks to my work in hyperbee. I can speak five languages including English, Albanian and German in professional proficiency, Turkish in a B1 level and french as a beginner

circuit  
design

programming  
in C. Python,  
Matlab

Zemax

Microsoft  
Excel

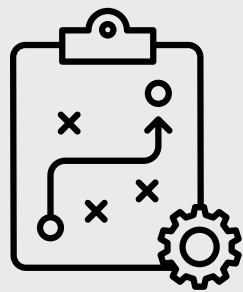
English,  
Albanian,  
German,  
Turkish,  
French

simulink  
simscape

# Skills Snapshot

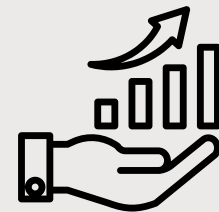
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HATIXHE MERKAJ



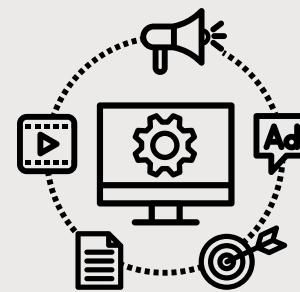
## Research

research and development



## Teamwork

thanks to my education background and experience in ITU hyperbee project team I can work in any environment and i can find a common language with anyone



Programming in C, Matlab and python, simulink , simscape, zemax optic studio simulation



## Analytics

Managment and organising skills

# Project

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HATIXHE MERKAJ

## Designing a Laser Filter

### Company

ITU EDL

### Role

My job was to do the simulation of the optical system in Zemax which is a ray simulation software. I was to make sure that the filter did what we wanted it to do without losing accuracy

### Goal

To build an optical system that will work as a laser filter that can be used for different wavelengths so that eye surgeons can use it for different surgeries in different parts of the eye

### Outcome

A good optical system was built and now everything is ready waiting for some last details from the company to finalize the project



# Project

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HATIXHE MERKAJ

Mathematically modelling the levitation system and designing the  
pid controller

## Company

ITU Hyperbee

## Goal

To perfectly model the levitation system of the vehicle we have built during the years. Find a suitable controller type design it and test it

## Role

Researching and understanding the system, extracting the mathematical model from simscape and running analysis in simulink and testing in real life in the vehicle with all the systems put together

## Outcome



The project is still  
continuing .

# Let's connect

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